

BLUE TEAM REPORT

HawkEye Reborn v9 Incident Investigation and Defensive Analysis

Report	Task 3 - Blue Team Report
Investigation	HawkEye Reborn v9 Incident Investigation and Defensive Analysis
Primary Methodology	Wireshark and DFIR methodologies
Incident Focus	Credential theft and SMTP-based exfiltration

Executive Summary

This report presents the findings of a forensic investigation conducted on a network packet capture associated with a HawkEye Reborn v9 malware infection. The objective of the investigation was to identify the infected host, determine the malware's behavior, analyze data exfiltration activities, and assess the impact on the organization. Using Wireshark and DFIR methodologies, the investigation revealed successful credential theft and SMTP-based exfiltration of sensitive information.

1. Incident Overview

During routine network analysis, suspicious SMTP traffic was identified originating from an internal host. Further investigation revealed that the host had been infected with HawkEye Reborn v9, an information-stealing malware capable of harvesting credentials and transmitting them to an attacker-controlled email account.

Affected Host	Value
Hostname	BEIJING-5CD1-PC
Internal IP Address	10.4.10.132
Public IP Address	173.66.146.112
Username	roman.mcguire

2. Investigation Methodology

Step 1: Packet Capture Analysis

The provided PCAP file was examined using Wireshark to identify suspicious network communications.

Step 2: Host Identification

Network traffic analysis was performed to determine the most active host within the environment.

Step 3: DNS Analysis

DNS queries were reviewed to identify suspicious domains contacted by the infected system.

Step 4: File Download Investigation

HTTP traffic was analyzed to identify malicious file downloads.

Step 5: SMTP Traffic Analysis

SMTP communications were inspected to identify credential exfiltration activities.

Step 6: Malware Attribution

Recovered evidence was used to determine the malware family and variant involved in the incident.

3. Key Findings

Finding	Value
Malware Identified	HawkEye Reborn v9
Malicious File	tkraw_Protected99.exe
Suspicious Domain	proforma-invoices.com
Resolved IP Address	217.182.138.150
Web Server Software	LiteSpeed
SMTP Server IP	23.229.162.69
SMTP Hostname	p3plcpnl0413.prod.phx3.secureserver.net
Attacker Email Account	sales.del@macwinlogistics.in
Exfiltration Interval	Every 10 minutes

4. Credential Theft Analysis

The malware successfully harvested sensitive credentials from multiple applications and services.

Account	Recovered Information
AOL Account	Username: roman.mcguire914@aol.com
Outlook Account	Email: roman.mcguire@pizzajukebox.com
Bank of America Account	Username: roman.mcguire Password: P@ssw0rd\$

The theft of banking and email credentials significantly increased the risk of account compromise and financial fraud.

5. Indicators of Compromise (IOCs)

Host Indicators

- BEIJING-5CD1-PC
- 10.4.10.132
- 173.66.146.112

Network Indicators

- proforma-invoices.com
- 217.182.138.150
- 23.229.162.69

Email Indicators

- sales.del@macwinlogistics.in

File Indicators

- tkraw_Protected99.exe

6. MITRE ATT&CK; Mapping

Tactic	Technique
Execution	T1204.002 – User Execution: Malicious File
Credential Access	T1056.001 – Keylogging
Credential Access	T1555 – Credentials from Password Stores
Discovery	T1082 – System Information Discovery
Collection	T1119 – Automated Collection
Exfiltration	T1048.003 – Exfiltration Over SMTP

7. Defensive Recommendations

- Deploy Endpoint Detection and Response (EDR) solutions to detect credential-stealing malware.
- Implement email filtering and attachment scanning to prevent delivery of malicious executables.
- Block unauthorized SMTP traffic from endpoints to external mail servers.
- Enable Multi-Factor Authentication (MFA) for email and banking services.
- Monitor DNS and SMTP traffic for unusual patterns and exfiltration attempts.
- Conduct regular security awareness training to reduce the risk of malicious file execution.

8. Lessons Learned

The investigation demonstrated how credential-stealing malware can use legitimate protocols such as SMTP to evade detection. Continuous monitoring of network traffic, endpoint activity, and authentication events is essential for identifying and responding to such threats.

9. Conclusion

The investigation confirmed that HawkEye Reborn v9 successfully infected the victim host, harvested sensitive credentials, and exfiltrated the collected data through SMTP communications. Timely detection and analysis enabled the identification of attacker infrastructure, compromised accounts, and key indicators of compromise, providing valuable intelligence for future defensive measures.